

INDUSTRY PROBLEM SOLVING TASK

Computer Vision – Method Selection and Constraint Analysis

1. Autonomous Vehicle Vision System Selection

Problem Requirements

The automobile company requires a vision method for an autonomous vehicle. The system must satisfy all three conditions: accuracy must be at least 90%, latency must not exceed 70 ms, and the computational requirement must be compatible with moderate resources.

Method	Accuracy	Latency	Computation
Viola-Jones	82%	30 ms	Low
YOLO	93%	60 ms	Moderate
Deep Learning	97%	90 ms	High

Method-wise Analysis

Viola-Jones has a latency of only 30 ms and requires low computation, so it is fast and resource-efficient. However, its accuracy is only 82%, which is below the required 90%. Therefore, it fails the accuracy constraint even though it satisfies the latency and computation constraints.

YOLO provides 93% accuracy, which is above the required minimum. Its latency is 60 ms, which is below the maximum acceptable value of 70 ms. It also requires moderate computation, matching the available vehicle resources. Therefore, YOLO satisfies all three constraints.

Deep Learning provides the highest accuracy at 97%, but its latency is 90 ms, which exceeds the 70 ms maximum. It also requires high computation while the vehicle supports only moderate resources. Therefore, Deep Learning fails both the latency and computational constraints.

Constraint Check

Method	Accuracy $\geq 90\%$?	Latency ≤ 70 ms?	Moderate computation compatible?	Decision
Viola-Jones	No	Yes	Yes	Reject
YOLO	Yes	Yes	Yes	Select
Deep Learning	Yes	No	No	Reject

Logical Evaluation

The problem uses an AND-type constraint: a method must satisfy accuracy, latency, and computational requirements simultaneously. Viola-Jones fails because $82\% < 90\%$. Deep Learning fails because $90\text{ ms} > 70\text{ ms}$ and its computation requirement is high. YOLO satisfies $93\% \geq 90\%$, $60\text{ ms} \leq 70\text{ ms}$, and moderate computation is supported.

Final Decision

YOLO is the only method that satisfies all the stated constraints. It provides 93% accuracy, 60 ms latency, and moderate computational requirements. Therefore, YOLO is the most suitable choice for the autonomous vehicle vision system. It provides a practical balance between reliable detection, real-time response, and available computational resources.

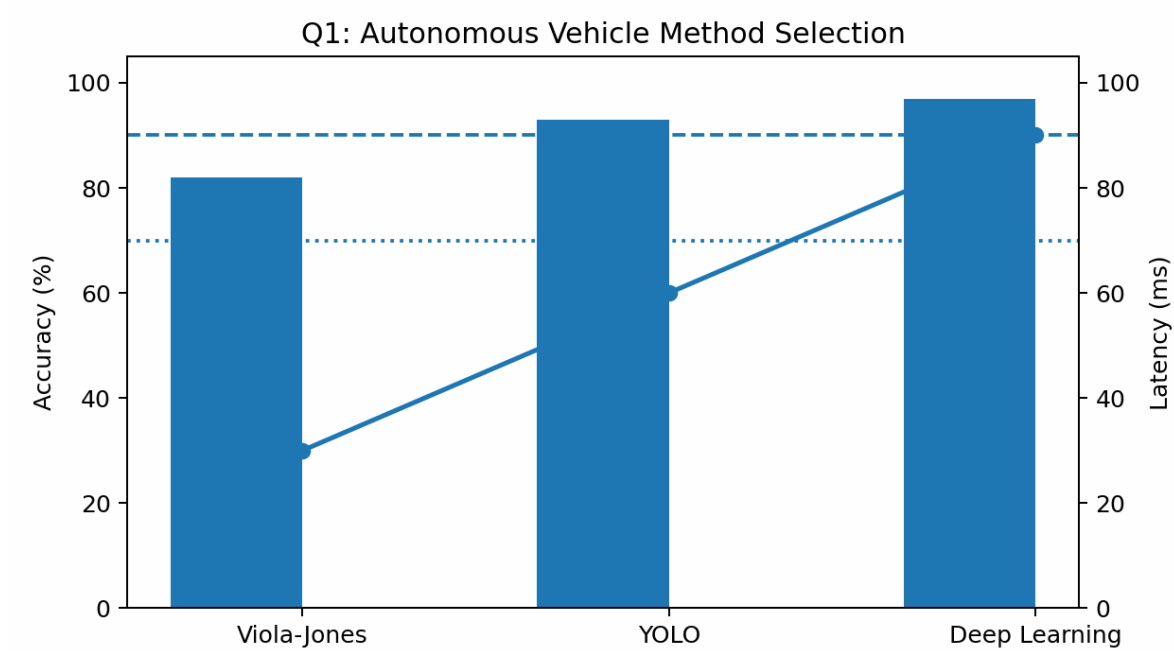


Figure 1: Accuracy and latency of the three autonomous-vehicle vision methods.

2. Smart Factory Product Inspection Selection

Problem Requirements

The manufacturing company needs a real-time computer vision system to detect defective products on a production line. Accuracy must be at least 95%, processing time must not exceed 50 ms, and the hardware supports moderate computation.

Method	Accuracy	Processing Time	Computation
Traditional Image Analysis	91%	25 ms	Low
YOLO	96%	45 ms	Moderate
Deep Learning	98%	75 ms	High

Method-wise Analysis

Traditional Image Analysis is very fast, with a processing time of 25 ms and low computational requirements. However, its accuracy is only 91%, which is below the required 95%. Since missing defective products can affect manufacturing quality, this method cannot satisfy the system requirement.

YOLO achieves 96% accuracy, exceeding the minimum requirement of 95%. Its processing time is 45 ms, which is below the maximum allowed 50 ms. It requires moderate computation, which matches the available hardware. Therefore, YOLO satisfies every constraint.

Deep Learning achieves the highest accuracy of 98%, which is desirable for defect detection. However, its processing time is 75 ms, exceeding the 50 ms limit. It also requires high computation, while only moderate computation is available. Thus, it cannot be selected under the stated deployment constraints.

Constraint Check

Method	Accuracy $\geq 95\%$?	Processing ≤ 50 ms?	Moderate computation compatible?	Decision
Traditional Image Analysis	No	Yes	Yes	Reject
YOLO	Yes	Yes	Yes	Select
Deep Learning	Yes	No	No	Reject

Logical Evaluation

The production system requires all constraints to be satisfied simultaneously. Traditional Image Analysis fails because $91\% < 95\%$. Deep Learning fails because $75\text{ ms} > 50\text{ ms}$ and it requires high computation. YOLO satisfies $96\% \geq 95\%$, $45\text{ ms} \leq 50\text{ ms}$, and moderate computation is supported.

Final Decision

YOLO is the only method that satisfies all the constraints for the smart factory. It provides 96% accuracy, 45 ms processing time, and moderate computational requirements. It therefore offers a suitable balance between defect-detection reliability, production-line speed, and hardware capability.

Practical Deployment Consideration

In a manufacturing environment, a missed defect can result in product quality problems, while excessive processing time can slow the production line. YOLO provides a 1-percentage-point accuracy margin above the required threshold and a 5 ms processing-time margin below the maximum limit. This gives a small but useful safety margin for deployment.

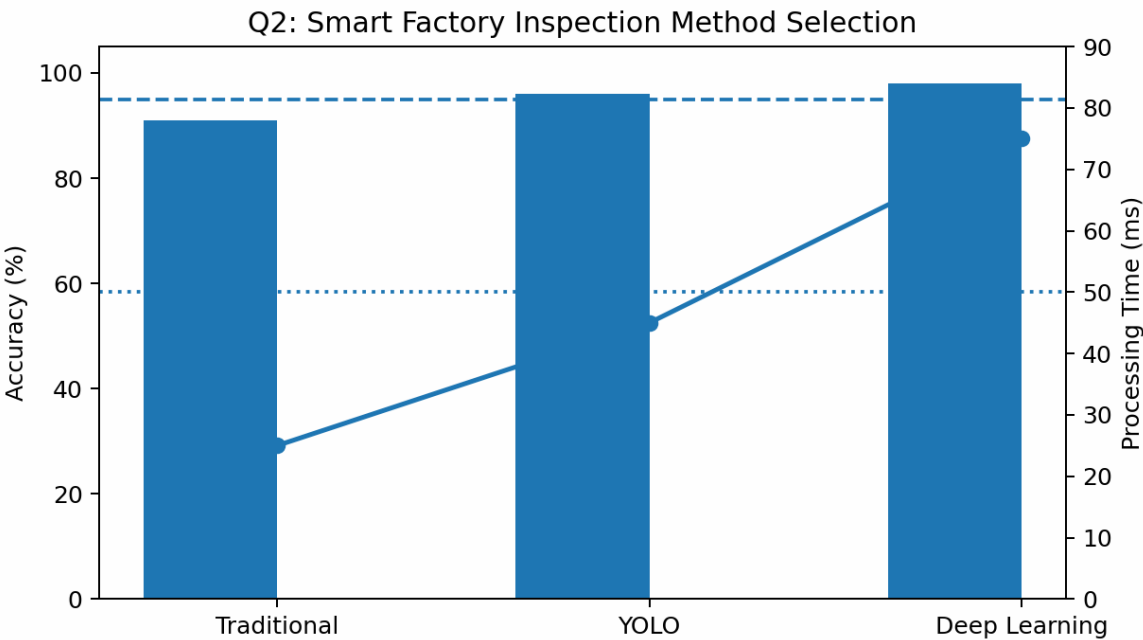


Figure 2: Accuracy and processing-time comparison for smart-factory inspection methods.

Overall Conclusion

In both industry problems, the correct decision is determined by satisfying every constraint rather than selecting the method with the highest accuracy alone. YOLO is selected in both cases because it meets the minimum accuracy requirement, remains within the latency or processing-time limit, and fits the moderate computational resources available.